

Lab 4: Loops

Computer Programming () |

Duration: 2 hours (in-lab) | **Topic:** while, do-while, for, nested loops, break/continue

Objectives

- Choose the correct loop form for the problem (counter-controlled vs. condition-controlled vs. at-least-once).
- Build nested loops to print patterns and search a 2D space.
- Use **break** and **continue** to control loop flow.
- Translate a mathematical sum or algorithm into an iterative program.

Quick Reference

Three loop forms.

```
while (cond) { /* tested first */ }
do { /* runs at least once */ } while (cond);
for (init; cond; update) { /* counter-controlled */ }
```

Loop control. **break** exits the nearest loop; **continue** skips to the next iteration of the nearest loop.

Common pattern: accumulator.

```
int sum = 0;
for (int i = 1; i <= n; i++) sum += i;
```

Common pattern: flag + early exit.

```
int found = 0;
for (int i = 2; i*i <= n; i++)
    if (n % i == 0) { found = 1; break; }
```

In-Lab Exercises (target: 2 hours)

In-Lab Exercise 1: Guess the Number (25 min)

Choose a secret number (e.g. 42) in your code. Repeatedly read a guess from the user with a **do-while** loop, printing **Too high**, **Too low**, or **Correct** as appropriate. After the user wins, print the number of guesses they used.

In-Lab Exercise 2: Prime Check with Early Exit (25 min)

Read a positive integer n and decide whether it is prime. Use a **for** loop from $i = 2$ to $\sqrt{n}$ and **break** as soon as a divisor is found. Handle $n \leq 1$ and $n = 2$ as special cases.

In-Lab Exercise 3: Pyramid Pattern (35 min)

Read an integer N and print a centered pyramid of asterisks with N rows. For $N = 5$:

```
  *
 ***
*****
```

```
*****
*****
```

Hint: row r (1-indexed) has $N - r$ leading spaces and $2r - 1$ asterisks.

In-Lab Exercise 4: Approximate e^x via Taylor Series (35 min)

Read x as a `double` and approximate $e^x = \sum_{n=0}^{\infty} x^n/n!$. Stop when the magnitude of the current term is below 10^{-6} . **Hint:** compute each next term incrementally as `term *= x / n`, where `n` is the new term index. Test with $x = 0, 1, 2$ and compare against `exp(x)` (link with `-lm`).

Take-Home (Paper Work). Solve the following on paper without using a computer or compiler. Hand-write your answers; submit at the start of the next lab. Goal: prepare you to dry-code during the written exam.

Trace & Predict 1: Off-by-one

What is the printed output?

```
#include <stdio.h>
int main(void) {
    int sum = 0;
    for (int i = 1; i < 5; i++) sum += i;
    printf("%d\n", sum);
    return 0;
}
```

Trace & Predict 2: Nested loop with continue

Predict the exact output:

```
#include <stdio.h>
int main(void) {
    for (int i = 1; i <= 3; i++) {
        for (int j = 1; j <= 3; j++) {
            if (i == j) continue;
            printf("(%d,%d) ", i, j);
        }
        printf("\n");
    }
    return 0;
}
```

Find the Bug 1: Infinite loop

Why does this program never terminate? Fix it (two reasonable fixes exist; show one).

```
#include <stdio.h>
int main(void) {
    int n;
```

```
scanf("%d", &n);  
while (n > 0);  
{  
    printf("%d\n", n);  
    n--;  
}  
return 0;  
}
```

Write Code on Paper 1: Sum and average of N inputs

Write a complete program that reads N followed by N integers, then prints their sum and their average (as a `double` with two decimal places).

Write Code on Paper 2: Multiplication table

Write a complete program that reads an integer N and prints the $N \times N$ multiplication table. Each cell must use a field width of 4 so the columns align.